

flyPad taxi route page

Ground > Taxi: the taxi route on the Navigraph airport map, from the gate to the runway holding point and from the runway exit to the gate, suggested and edited to the ATC clearance

Branch	pr/26-efb-taxi-route
Aircraft	A32NX and A380X (flyPad)
Commits	b97742c40 feat(efb): taxi route page on the Navigraph airport map
Base	PR 23 (landing calculator: the shared runway exit module btwExits.ts)
Status	CI checks pass; merged into develop 9cd78dd05; tested in the sim on 2026-09-28 at RJTT (user-confirmed).

Purpose

The A380 OANS shows the airport map but no taxi route: the 2011 A380 FCOM has no taxi routing (the crew marks the map with flags and crosses, DSC-34-10-70-20) and MSFS has no D-TAXI. The flyPad page gives the crew the route of the ATC taxi clearance on the same Navigraph airport map (AMDB) the OANS uses, with the runway crossings, so the cockpit displays stay true to the FCOM.

Specification

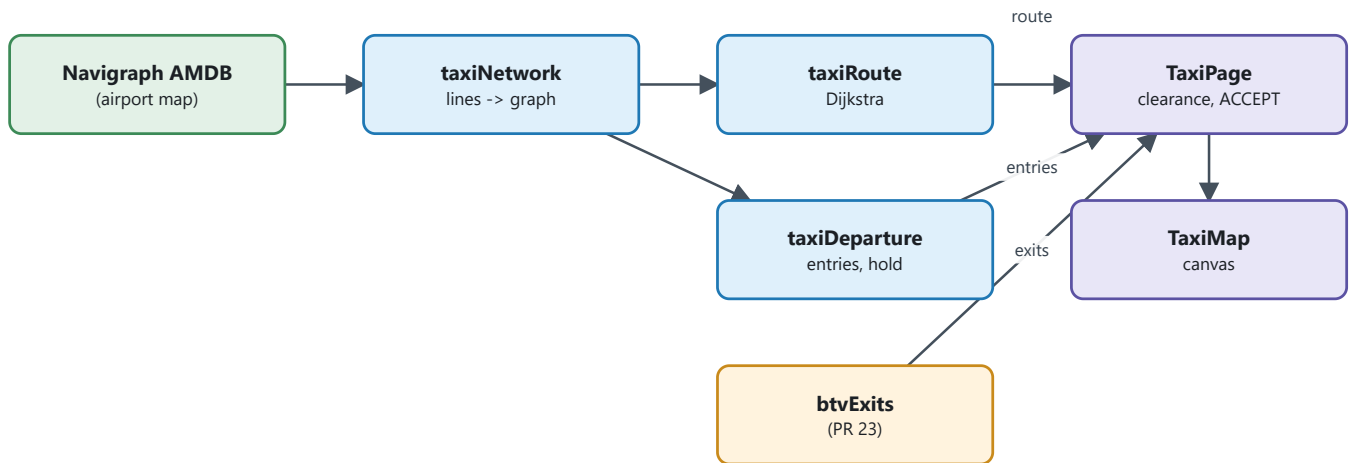
Item	Behaviour
Departure (default)	from a gate / ramp / cargo stand, or the aircraft position, to a runway entry (the first one = full length, the others with the runway length remaining); the route stops at the runway holding position line crossed last (HOLD RWY stop bar)
Arrival	from a runway exit (the exits of the landing calculator) or the aircraft position to a stand
Suggestion	the shortest route along the taxi lines; SUGGEST copies its taxiways into the clearance field
ATC clearance	the taxiways typed in order: the route follows them strictly (unnamed connecting lines allowed, the runway entry line after the last one); ACCEPT draws it solid; an impossible clearance is shown in red
Map	aprons, taxiways, runways, guidance lines, stands (tap one to choose it), taxiway names, runway entries, route with its taxiway names, runway crossings in red, aircraft symbol; drag, zoom, fit
Stand names	typed with or without leading zeros and separators ("16" finds "016"); suggestions list the names that start with, then contain, the typed text

Design

- Shared module `fbw-common/src/systems/shared/src/taxi/`: `taxiNetwork.ts` builds the network from the AMDB guidance lines (taxiway, runway exit and stand guidance lines; runway centrelines only split the lines that cross them and are never followed); lines split where they cross, line ends within 3 m joined (T junctions), a dead end of a disconnected part joined to the nearest line of another part within 15 m (gaps in the airport map).
- `taxiRoute.ts`: Dijkstra over (node, index of the cleared taxiway) states; unnamed lines cost x1.2; the lead-in lines of other stands and other exits x3 as a last resort; from the aircraft, a 2000 m penalty for starting against its heading. `taxiRouteRunwayCrossings` samples the route every 5 m in the runway polygons.
- `taxiDeparture.ts`: the runway entries (lines leaving the runway area within 40 m of the centreline, unnamed lead-on lines named after the taxiway they come from) and the cut at the holding point (last holding position line crossed within the last 400 m).
- `taxiAmdb.ts`: lines, stands, runways, runway axes, holding lines and map shapes from the AMDB features; stand name matching.

- flyPad: `Ground/Pages/Taxi` (TaxiAirport loads the AMDB layers through the Navigraph client, cache of 3 airports; TaxiMap canvas drawn larger than the view so a drag only moves it; TaxiPage), store slice `taxiRoute` (kept across the flyPad pages), Taxi tab in Ground.

 Component added or changed by the feature
 MSFS / simulator
 Aircraft system (systems host, FMS, FWS)
 Display / page
 Data, LVar, storage
 Crew, flypad, external program



Architecture

Tests

9 unit tests in `taxi/taxiRoute.spec.ts` on a synthetic airport: suggestion and clearance routes, clearance errors, runway crossings, the runway entries (full length, crossing taxiway, unnamed lead-on line), the holding point cut, an intersection departure, stand name matching, the join of a lead-in line that stops short of the taxi line; plus the runway exit start points in `btvExits.spec.ts`.

Limitations

- Needs a Navigraph account (the AMDB); without it the page says so.
- The route follows the guidance lines of the map; one-way taxiways, closures and NOTAMs are not known.
- The route is shown on the flyPad only, not on the OANS (not in the FCOM).